



Kshetra Email Integration Requirements

Free Implementation — Gmail API & OAuth 2.0

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Project:	Kshetra Field Sales App	Status:	Awaiting Team Confirmation

Team 102, please read the following requirements carefully. We are planning to integrate a complete email functionality into our Kshetra Field Sales App using free software and APIs.

The objective is to enable field employees to communicate with customers directly through Kshetra using their existing Gmail accounts, without purchasing Google Workspace, Microsoft 365, or any paid email service.

All development must be properly integrated with our existing Kshetra application, customer database, employee authentication, and field visit modules.

1. TECHNOLOGY AND SOFTWARE REQUIREMENTS

Use the following free technologies for implementation:

- **Gmail API:** For sending real emails through employees' existing Gmail accounts, subject to Google's API quotas and OAuth requirements.
- **Google OAuth 2.0:** For securely connecting employees' Gmail accounts without collecting their passwords.
- **Mailpit:** For local development and testing of email functionality without sending actual emails to customers.
- **Node.js / Express:** For backend email processing and Gmail API integration.
- **React:** For the email interface and customer communication module.
- **PostgreSQL:** For storing email logs, connection metadata, and customer communication history.

Do not integrate paid email delivery services, paid SMTP providers, Google Workspace subscriptions, or Microsoft 365 subscriptions for this implementation. Use existing Gmail accounts for real-email testing, subject to Google's account policies, OAuth restrictions, and sending quotas.

2. GMAIL ACCOUNT INTEGRATION

Implement secure Gmail account integration using OAuth 2.0. Required functionality:

1. Employees should be able to connect their existing Gmail accounts through a "Connect Gmail" option inside Kshetra.
2. Use Google's official authorization and consent process.
3. Employees must be able to connect and disconnect their accounts.
4. Store access and refresh tokens securely on the backend, using appropriate encryption.
5. Never request or store employee Gmail passwords.
6. Use the minimum necessary Gmail API permissions, starting with `gmail.send` for sending emails.
7. Implement secure token refresh, authorization validation, and error handling.

Initially, focus on sending-only functionality. Do not request full inbox access unless it is required for a separately approved feature.



Please ensure that Google's OAuth testing restrictions, verification requirements, and applicable Gmail sending limits are considered before production deployment.

3. CUSTOMER EMAIL FUNCTIONALITY

Add a "Send Email" option inside customer profiles and field visit records. The email composer must support:

- Recipient email address automatically populated from the customer database.
- CC, subject, and message body.
- Editable email content before sending.
- Attachments such as quotations, product brochures, PDFs, and authorized field visit photographs.
- Sending emails through the employee's connected Gmail account.

The email should be sent from the connected employee's Gmail account, subject to Google's account and sending policies. Employees should not need to leave Kshetra and manually copy customer details into Gmail.

4. AUTOMATIC FIELD VISIT REPORTS

Integrate email functionality with the existing field visit submission workflow. When an employee submits a field visit, Kshetra should provide the option to email the visit report to the assigned manager or designated team. The report should contain:

- Employee name and visit ID.
- Customer details and visit date.
- Visit location, where appropriate and authorized.
- Visit notes and follow-up date.
- Relevant photographs or a generated PDF report.

The manager or report recipient must be configurable according to company requirements. Initially, implement this as a configurable feature, allowing the employee or authorized workflow to initiate the report email. Automatic sending rules can be added after testing.

5. EMAIL HISTORY AND DATABASE INTEGRATION

Create the necessary database tables or migrations to maintain email history. Each email record should include:

- Employee ID and customer ID.
- Associated field visit ID.
- Sender and recipient email addresses.
- Email subject and timestamp.
- Provider message ID, where available.
- Email status, such as queued, sent, failed, or delivered, wherever supported.

The system must associate each email with the correct customer and field visit. Employees should be able to view their sent email history inside Kshetra.

Do not mark an email as delivered merely because Gmail accepted the send request. Distinguish between a successful API submission and confirmed delivery.

6. FREE EMAIL TESTING USING MAILPIT



Before connecting real Gmail accounts, use Mailpit for local development and testing. Mailpit must be used to test:

- Email composer functionality.
- Email templates and formatting.
- Field visit report generation.
- PDF and attachment handling.
- Backend email processing and error handling.

Mailpit captures test emails locally without delivering them to actual customers. Once the functionality is working correctly in the local environment, proceed with Gmail API integration and real-email testing using an authorized Gmail account.

No paid email delivery service should be required for the local testing phase.

7. TECHNICAL ARCHITECTURE

Please organize the implementation into separate modules:

Frontend	Backend	Database
• Gmail connection and disconnection interface. • Customer email composer. • Field visit report email option. • Sent email history and status display.	• Google OAuth authentication and callback handling. • Gmail API integration and message sending. • Attachment handling and validation. • Email logging, retries, and error handling. • Authorization checks and token management.	• Email connection metadata. • Email logs. • Customer and visit associations. • Attachment references, where required.

Keep Gmail-specific functionality inside a separate email service or provider adapter so that the existing Kshetra modules remain maintainable. All Gmail API credentials and OAuth secrets must remain on the backend.

8. SECURITY AND ACCESS CONTROL

Since Kshetra contains customer information, location data, and field photographs, proper security is mandatory.

- Employees must only be able to send emails using their own authorized connected accounts.
- Prevent unauthorized access to another employee's mailbox.
- Validate customer email addresses and attachment permissions.
- Protect access and refresh tokens through secure storage and encryption.
- Implement proper session validation and backend authorization.
- Avoid exposing sensitive customer information in logs or unauthorized email attachments.



9. DEVELOPMENT PHASES

Phase 1 — Core Email Functionality	Phase 2 — Advanced Functionality
1. Mailpit setup and local email testing. 2. Gmail OAuth account connection. 3. Customer email composer and Gmail API sending. 4. Attachments and field visit report emails. 5. Email history and database integration. 6. Security checks and deployment testing.	1. Gmail inbox synchronization. 2. Receiving and viewing customer replies inside Kshetra. 3. Reply and forward functionality. 4. Customer-wise email conversation history. 5. Follow-up reminders and notifications.

Complete and test Phase 1 before proceeding to Phase 2. Any additional Gmail permissions required for inbox access must be reviewed separately.

10. REQUIRED TEAM UPDATE AND DELIVERABLES

Before starting development, please provide a clear response covering the following:

1. Confirm the existing Kshetra frontend, backend, and database architecture.
2. Confirm how Gmail API and OAuth will be integrated.
3. Identify the required Google Cloud configuration and OAuth approvals.
4. Provide the task-wise responsibility allocation for each team member.
5. Provide an unambiguous development start date and Phase 1 completion date.
6. Identify any access, configuration, or approval required from our side.

The final deliverable must include working source code, database migrations, configuration documentation, testing evidence, and a properly functioning integration in the deployed Kshetra environment.

Important: Our requirement is to implement this using free software and existing Gmail accounts wherever possible. Do not introduce paid subscriptions or paid email service dependencies without prior approval.

Please acknowledge this requirement, review the complete scope, and submit your implementation plan with clearly assigned responsibilities and deadlines.

— Anand Srivastava

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